

SHORT BIO. My research focuses on causal decision making (NeurIPS-25, Preprint, ICLR-26) and causal effect identification (NeurIPS-24). I investigate how to formally model diverse real-world problems within a causal framework and derive optimal decisions from such structured representations. In particular, I develop methods for robust decision-making in settings under environmental uncertainty (NeurIPS-25), domain shifts between training and deployment environments (Preprint), and even scenarios involving counterfactual actions (ICLR-26).

EDUCATIONS

Graduate School of Data Science, Seoul National University

Integrated M.S. and Ph.D. in Data Science

Advisor: Prof. Sanghack Lee.

Overall GPA: 4.02 /4.3.

Seoul, Republic of Korea

Sep 2022 - Aug 2027 (expected)

Department of Mathematics and Statistics, Sejong University

B.S. in Applied statistics and Mathematics (Double major), *Cum Laude*

Advisors: Prof. Ji Eun Lee and Jin Hee Yoon.

Overall GPA: 4.04/4.5; Major GPA: 4.46 (Applied Statistics), 4.38 (Mathematics).

Seoul, Republic of Korea

Mar 2016 - Aug 2022

PUBLICATIONS

* for joint first authorship and † for corresponding author.

Min Woo Park and Sanghack Lee†. [Counterfactual Structural Causal Bandits](#). *International Conference on Learning Representations (ICLR)*, 2026.

Min Woo Park, Andy Arditi, Elias Bareinboim†, Sanghack Lee†. [Structural Causal Bandits under Markov Equivalence](#). *Neural Information Processing Systems (NeurIPS)*, 2025, Columbia CausalAI Laboratory, Technical Report (R-122)

Min Woo Park and Sanghack Lee†. [On Transportability for Structural Causal Bandits](#). *arXiv preprint*, 2025.

Yonghan Jung†, Min Woo Park, and Sanghack Lee†. [Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference](#). *Neural Information Processing Systems (NeurIPS)*, 2024.

Min Woo Park*, and Ji Eun Lee*†. [Computation of the iterated Aluthge, Duggal, and Mean transforms](#). *Filomat*, 2023.

Min Woo Park*, Taehui yun*, Youngin Jang, Younsuk Yeom, Jonghwan Kim, LG AI Research, Sanghack Lee†. [Breaking Bad: Component-Wise Parent Deletion for Score-Based Causal Discovery](#). under review.

Ji Eun Lee*† and Min Woo Park*. [Convergence of the iterated mean transforms of a \$2 \times 2\$ matrix](#). under review.

AWARDS AND HONORS

Graduate Student Instructor (GSI) scholarship | Seoul National University Spring 2024, Spring 2025, Spring 2026

BK21 scholarship Fall 2024, Fall 2025

First place merit scholarship | Sejong University Fall 2020, Fall 2021

Fourth place award | The 9th College of Natural Sciences Academic Conference, Sejong University Nov 2021

▷ Mitigating spurious regression in time-series data via Granger causality.

Third place award | The 8th College of Natural Sciences Academic Conference, Sejong University Nov 2020

▷ The significance of the Euler's number and its applications.

ACADEMIC SERVICES

Conference Reviewer: 2026 ICLR, ICML, 2025 ICML

PROJECTS

Towards More Scalable Score-based Causal Discovery LG AI Research	Sep 2025 - Aug 2026
▷ Working on developing scalable score-based causal discovery algorithms.	
Hyundai NGV Data Boot-Campus Hyundai Motors	July 2025 - Sep 2025
▷ Development of high-voltage battery performance prediction technology.	
Hyundai NGV Data Boot-Campus Hyundai Motors	July 2024 - Sep 2024
▷ Development of an intelligent system for predicting vulnerable regions of automotive body panel stiffness through differential geometry-based automatic curvature analysis.	

TEACHING ASSISTENTS

Causal Inference for Data Science	Spring 2024, Fall 2024, Spring 2025, Spring 2026
▷ Bayesian Networks and Pearl's Graphical Causal Inference Framework.	
Machine Learning and Deep Learning for Data Science II	Fall 2023
▷ Kernel Methods, Gaussian Processes, and Related Topics for Machine Learning	
Basic Mathematics for Artificial Intelligence	Spring 2022
▷ Linear Algebra, Basic Statistics, and Probability Theory for Artificial Intelligence.	

PROFESSIONAL TALKS AND POSTERS

Counterfactual Structural Causal Bandits

ICLR 2026 poster session Rio de Janeiro, Brazil	Apr 2026 (expected)
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Structural Causal Bandits under Markov Equivalence

ExploreCSR poster session Seoul, Republic of Korea	Dec 2025
NeurIPS 2025 poster session San Diego, USA	Dec 2025
JKAIA 2025, NeurIPS preview session Seoul, Republic of Korea	Nov 2025
SNU GSDS student research seminar Seoul, Republic of Korea	Apr 2025

Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference

NeurIPS 2024 poster session Vancouver, Canada	Dec 2024
SNU GSDS student research seminar Seoul, Republic of Korea	Dec 2024

REFERENCE

Sanghack Lee | Seoul National University

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Graduate School of Data Science

Associate professor

Ji Eun Lee | Sejong University

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Department of Mathematics and Statistics

Associate Professor

Jin Hee Yoon | Sejong University

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Department of Artificial Intelligence and Information Technology

Associate Professor

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